Smart Quotation Generator - Project Report

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1. Introduction

The Smart Quotation Generator is a Java application that lets a user prepare a price quotation through a desktop-style form and have it handled by a backend server. It has two parts: a Java Applet frontend that the user interacts with, and a Java server that processes requests on the local machine.

Objective: to build a simple, working client-server quotation tool in core Java, using an Applet for the interface and a Java server for the logic.

2. Technology stack

The whole project is written in core Java, with an HTML page used only as the launcher for the applet.

Layer	Technology	Role
Frontend	Java Applet (`QuotationApplet`)	The form and screen the user works with
Backend	Java server (`QuotationServer`)	Listens on port 8080 and handles the applet's requests
Launcher	HTML page (`index.html`)	Loads the applet and passes it the server address
Runtime	JDK 8 to 19 with `appletviewer`	Runs the applet, since browsers no longer do

4. Project files

File	Purpose
`index.html`	Launch page; embeds the applet at 760 x 520 and sets the server address to `http://localhost:8080`
`QuotationApp.java`	Client-side source for the applet interface

File	Purpose
`QuotationServer.java`	Backend server source
`run.bat`	Windows script that compiles and starts the project
`golive.cmd`	Same one-click start for Windows
`golive.command`	Same one-click start for macOS
`README.md`	Short setup notes

5. How to run

Install JDK 17 first; `appletviewer` was removed in JDK 20 and later, so newer versions cannot run the project.

1. Open a terminal in the project folder.
2. Compile everything with `javac *.java`.
3. Start the server with `java QuotationServer` and leave it running.
4. In a second terminal, run `appletviewer http://localhost:8080/index.html`.

The applet opens in its own window, not in Chrome or Safari. Running `golive.cmd` (Windows) or `golive.command` (macOS) does all four steps in one go.

6. Limitations, future scope and conclusion

Limitations

- Java Applets are deprecated and no browser can run them, so the app works only through `appletviewer` on JDK 8 to 19.
- The server address is fixed to `localhost`, so it runs on one machine only.

Future scope

- Replace the applet with a Swing, JavaFX or web frontend so it runs on current Java and in a browser.
- Add saving of past quotations and export to PDF.

Conclusion

The project shows a complete client-server design in core Java: an applet frontend, a server backend, and a launch script that ties them together. Its main constraint is the age of the applet technology, which the future-scope items address.